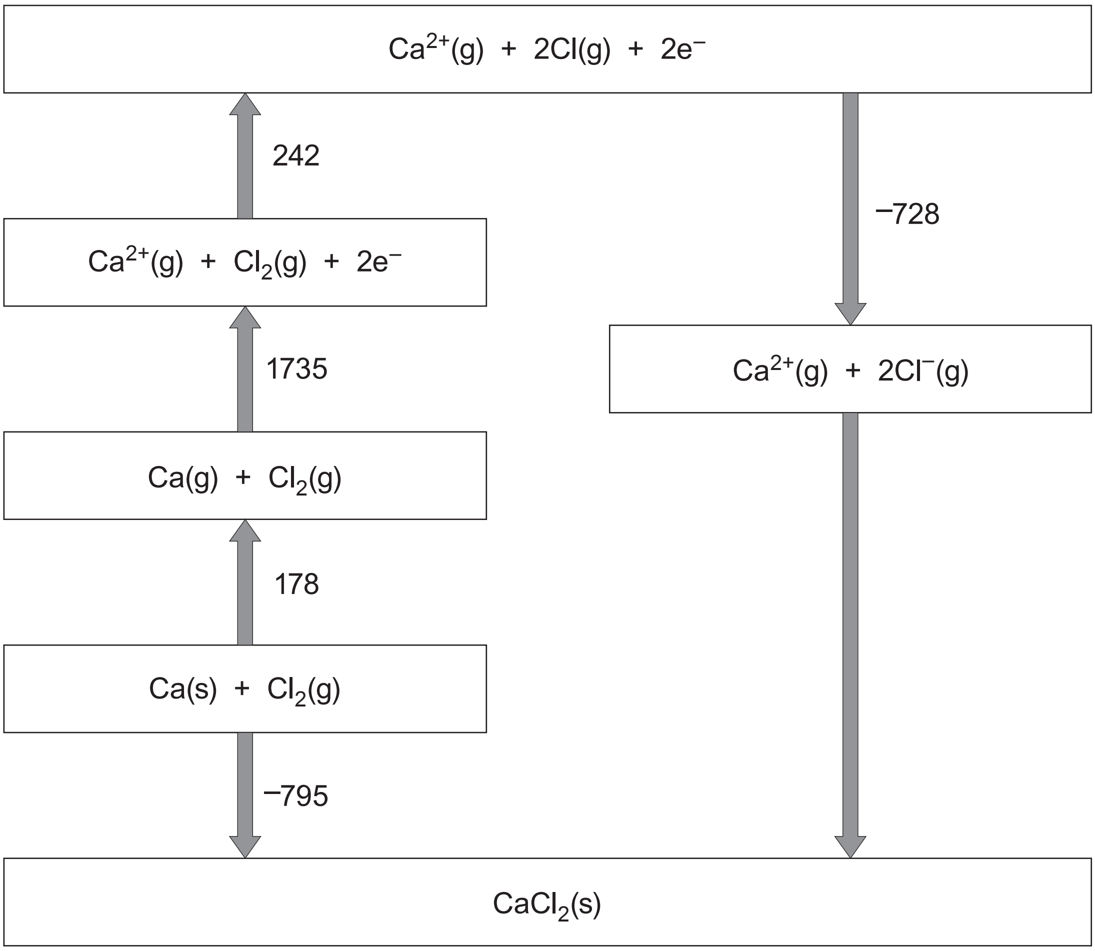
SECTION B

*Answer **all** questions in the spaces provided.*

9. Calcium chloride is an ionic solid that is used as a drying agent due to its ability to absorb water and form a range of hydrated crystals.

(a) The Born-Haber cycle below shows the formation of calcium chloride from its elements.

All values shown are standard enthalpy changes and are given in $kJ\,mol^{-1}$.



Examiner only

(i) Give the value of the standard enthalpy change of atomisation of chlorine. [1]

.... kJ mol^{-1}

(ii) Calculate the standard enthalpy of lattice formation of $CaCl_2$. [2]

$\Delta_{latt\ form}H^{\theta}$ = ... kJ mol^{-1}

(iii) A student incorrectly states that the second ionisation energy of calcium is 1735 kJ mol^{-1} as this is the energy needed to form Ca^{2+}(g) in the Born-Haber cycle.

Suggest a possible value for the second ionisation energy of calcium, giving a reason for your answer. [2]

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(b) The reaction occurring when calcium chloride is used as a drying agent is shown below.

$$CaCl_2 + xH_2O \longrightarrow CaCl_2.xH_2O$$

(i) A laboratory manual suggests a **minimum** temperature of 200 °C to remove all the water from the hydrated calcium chloride. At this temperature the reaction occurring is as follows.

$$CaCl_2.xH_2O \longrightarrow CaCl_2 + xH_2O$$

A student calculates the enthalpy change when calcium chloride becomes dehydrated as 78.0 kJ mol^{-1}.

Assuming that this data is correct, calculate a value for the entropy change when calcium chloride becomes dehydrated at this temperature. [3]

ΔS = ... J K^{-1} mol^{-1}

(ii) In the drying process 5.20 g of hydrated calcium chloride forms 3.15 g of anhydrous calcium chloride.

Calculate the value of x in the original $CaCl_2.xH_2O$. [3]

x = ...

(iii) State the colour observed in a flame test using calcium chloride. [1]

....

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(c) Calcium halides react with concentrated sulfuric acid in a similar manner to sodium halides.

State the observation(s) when concentrated sulfuric acid is added to calcium chloride and calcium bromide. Explain why the observations are different. [3]

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